RAFAEL M. CHAVEZ

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Education

Georgia Institute of Technology — Atlanta, GA

B.S. Mechanical Engineering, Minor in Applications of AI/Machine Learning | GPA: 3.9 — *Aug. 2023 – May 2027*

Work Experience

Autonomy and Controls Engineering Intern | *Sea Machines Robotics* — May 2026 – Nov. 2026

- Tested and evaluated radar sensor hardware to ensure reliable performance for autonomous vessel deployments.
- Built a streamlined data ingestion application supporting live mission data capture for downstream analysis.
- Developed a vessel-agnostic dynamic model parameter estimation tool achieving sub-5% average accuracy in simulation, supporting collision avoidance, fuel optimization, and trajectory planning.
- Applied digital signal filtering techniques to improve fidelity of parameter state estimation model.
- Validated estimated hydrodynamic parameters using motion model simulations with injected Markov-Gauss sensor noise and physical environmental wave disturbances.

Process Engineering Intern | *Amazon Robotics* — May 2025 – Aug. 2025

- Developed cycle time studies and SOP's to define discrete scopes of work across manufacturing operations.
- Cut part delivery time 81%, increased discrete materials per cycle 18x, and reduced station headcount 60%.
- Drove 85% increase in technician task efficiency through Pareto analysis.
- Identified bottlenecks via time cycle analysis, cutting manufacturing downtime and reducing headcount by 22%.

Research Experience

Project Lead and Research Assistant | *Institute for Robotics and Intelligent Machines* — Dec. 2025 – Present

- Deployed GPT-4o and OpenCV to detect, locate, and manipulate objects with UR5 and Rainbow robot arms.
- Trained an RF-DETR detection model to improve end effector relative object manipulation and orientation estimation.
- Built a $50,000 cost-saving machine vision part inspection prototype to compete in the TE Connectivity AI Cup.

Research Assistant | *Exoskeleton & Prosthetic Intelligent Controls Lab* — Aug. 2024 – May 2025

- Engineered hardware for a hip exoskeleton assistive device targeting stroke patient rehabilitation.
- Analyzed exoskeleton biomechanics via motion capture and simulation to inform design decisions.
- Cut hardware housing fixture mass 55% and volume 32% while optimizing cable routing and component cooling.
- Designed parts in SolidWorks and manufactured using 3D printing and pre-impregnated carbon fiber layering.

Publications

R. Chavez, "Edge AI Vision Model for Catheter Shaft Inspection," *IEOM 2026 World Congress*, Irvine, CA, accepted for publication, Oct. 2026.

Projects

Autonomous Chess-Playing Robot | *Personal Project* — Aug. 2026 – Present

- Designing and building a chess-playing robot arm from scratch, including custom 3D-printed linkages, hardware architecture, and integrated motors, sensors, and microcontroller.
- Developing a computer vision model to detect board state and piece positions in real time, and integrating a Stockfish API to compute optimal moves and drive physical arm motion.

Mobile Robot Navigation | *Robotics Coursework* — Jan. 2026 – May 2026

- Built forward/inverse kinematic models in MATLAB, optimizing trajectory planning and closed-loop end-effector control.
- Programmed TurtleBot3 autonomous navigation in ROS, fusing LiDAR, IMU's and odometry for path following.
- Trained CNN's for robotic object classification, tuning architectures and hyperparameters to maximize accuracy.

Leadership

Director of Corporate Relations | *Pi Tau Sigma Mechanical Engineering Honor Society* — Jul. 2025 – May 2026

Student Advisory Board Member | *Woodruff School of Mechanical Engineering* — Mar. 2025 – May 2026

Undergraduate Teaching Assistant and Mentor | *GT Tutoring and Academic Support* — Jan. 2024 – Dec. 2024

Technical Skills

Relevant Coursework: Robotics, Mechatronics, Machine Learning, Circuit Design, Machine Design, System Dynamics
Programming & Software: MATLAB, Python, C, C++, MOOS-IvP, Excel, Linux, WSL, Git
Robotics & AI: ROS, OpenCV, YOLO, Deep Learning, CNNs, Neo4j, Cypher, RoboFlow
CAD & Design: SolidWorks, Autodesk Fusion 360
Manufacturing & Methods: 3D Printing, Machining, Soldering, Carbon Fiber Layering, Six Sigma, 5S